

ISOS abstract

Towards the investigation of the metastable behavior of perovskite samples using indoor light cycling techniques and outdoor assessment

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Several perovskite-based modules have been tested outdoors and indoors using a light-dark cycling approach at different temperature levels with the aim of comparing the electrical parameters evolution between samples tested indoors with samples tested outdoors at different environmental conditions. Particularly, indoor light-dark measurements at fixed irradiance levels (1 sun), high/low temperature levels and varying irradiance that mimic the outdoor irradiance profile have been implemented in this work. The cycled metastability behavior of the samples was compared using the different cycling approaches indoors and then compared with the outdoor metastability trend of the samples at different test locations in Germany and Cyprus. Open-circuit bias load was applied at all samples under test indoors and outdoors for a reliable comparison between the results.

Comparison of the power output in samples tested indoors with cycling testing at fixed irradiance and different temperature levels showed that the output power at low temperatures presented more pronounced light soaking effects and minimum degradation compared to samples tested at higher temperatures in agreement with outdoor results. The same trend (higher light soaking, no degradation) was obtained in open-circuit voltage and Fill Factor parameters in samples tested at low temperature cycling tests. Power output degradation and recovery during light/dark cycling respectively were found to be statistically higher in samples tested at higher temperatures in agreement with past measurements.

So far, results have demonstrated similar cycled behavior of the major electrical parameters in indoor tests performed at fixed irradiance levels and different temperature levels especially during the last cycle testing. During the first cycles, a larger impact of

light soaking is found on samples tested at lower temperatures. The tests have also shown that the strength of metastable behavior outdoors in samples tested in Cyprus differs considerably compared to that obtained indoors at fixed irradiance levels. Furthermore, larger metastability in power output was obtained in Cyprus test site compared to Germany giving evidence that the metastable behavior of perovskite samples is strongly correlated to the test location. Outdoor results also demonstrated opposite behavior of the performance degradation and recovery over the day/night cycles in Cyprus and in Germany.

By studying the performance evolution over the day outdoors in Cyprus during summer months this was found to be determined by series resistance losses attributed to the different irradiance levels over the day thus affecting the Fill Factor and performance behavior. This trend was not found in indoor cycling testing at fixed irradiance levels giving evidence that the electrical parameters behavior in cycling testing at fixed irradiance levels cannot reproduce the outdoor data collected under continuous changing of the environmental conditions.

More indoor and outdoor tests are underway and are expected to shed more light on the metastable behavior of perovskite samples outdoors and indoors at different dark/light cycling approaches.

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